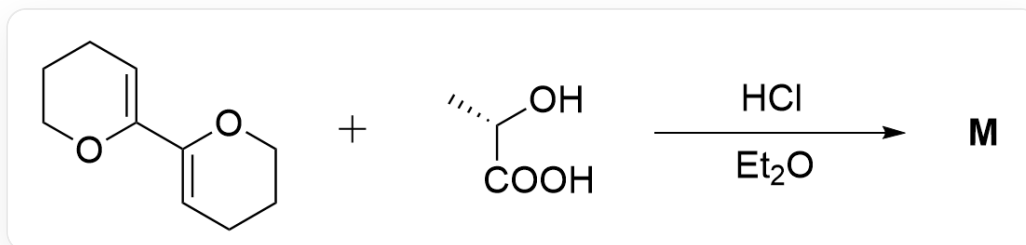


Question

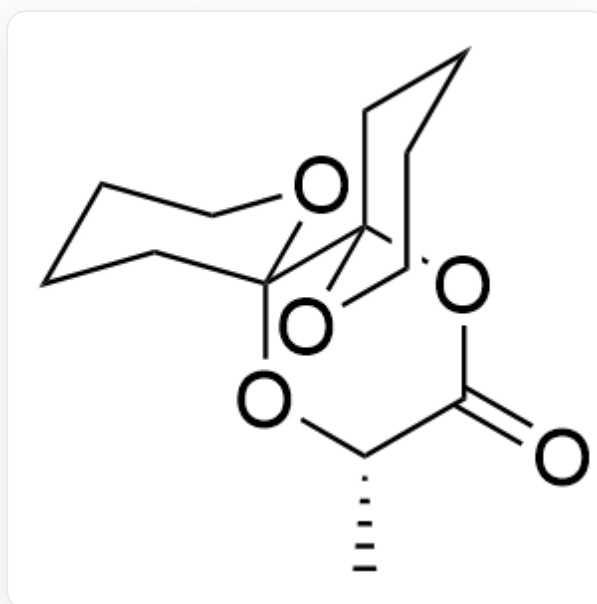


O[C@@H](C)C(O)=O.C1(C2=CCCCO2)=CCCCO1>CCOCC.Cl>[M], M is the reaction product

When the two substrates shown in the figure are placed in an ether solution of HCl, the reaction mainly yields the tricyclic product **M**. Without considering enantiomers, give the structural formula of **M**.

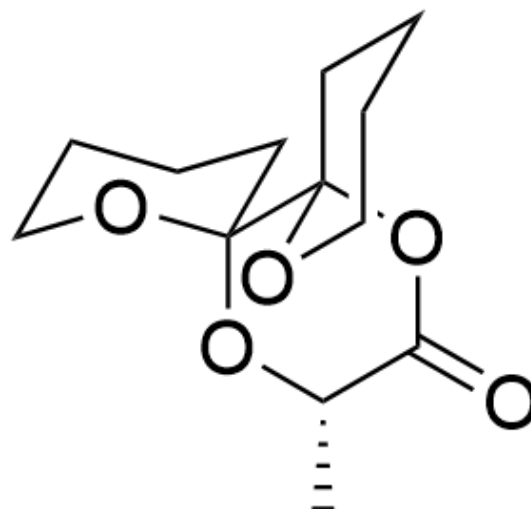
A. All other options are incorrect

B.



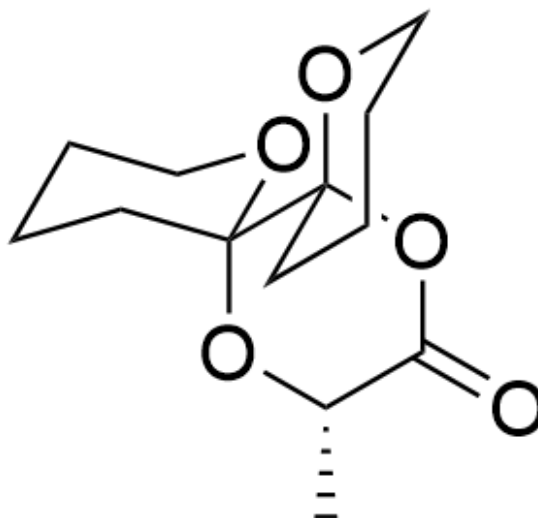
O=C(O[C@]12OCCCC2)[C@H](C)O[C@]31CCCCO3

C.



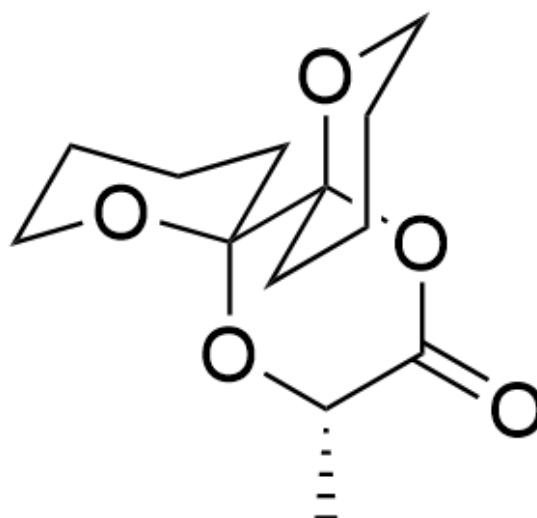
O=C(O[C@]12OCCCC2)[C@H](C)O[C@]31OCCCC3

D.



O=C(O[C@]12CCCCO2)[C@H](C)O[C@]31CCCCO3

E.



O=C(O[C@]12CCCCO2)[C@H](C)O[C@]31OCCCC3

Answer

Correct Answer: **B**

Detailed Explanation

Acid-catalyzed, this condensation reaction is reversible, so it is speculated that **M** exists in its most stable form.

CHECKPOINT

1 PTS

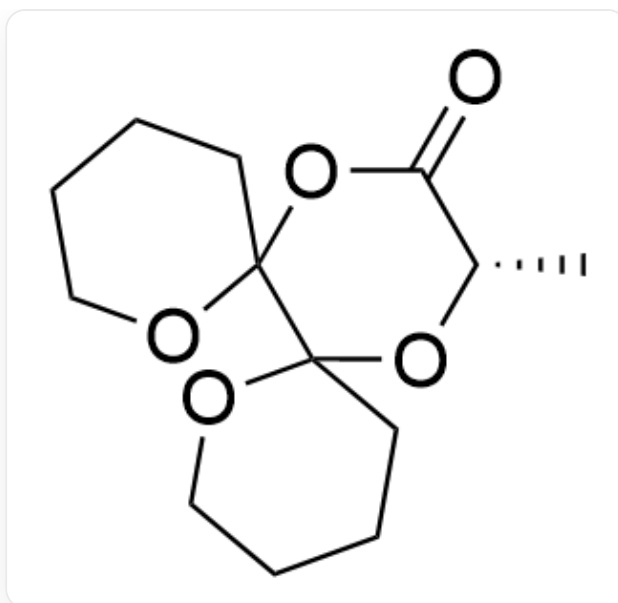
Acid-catalyzed, this condensation reaction is reversible, so it is speculated that **M** exists in its most stable form.

First, according to the hint of the tricyclic product, it is easy for us to think that this ketal structure can be formed through a simple two-step addition reaction.

CHECKPOINT

1 PTS

First, according to the hint of the tricyclic product, it is easy for us to think that this ketal structure can be formed through a simple two-step addition reaction.



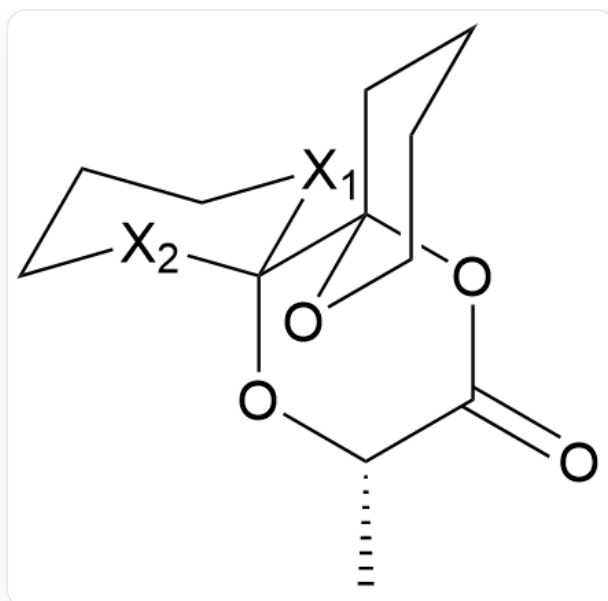
C[C@@H]1OC2(OCCCC2)C3(OC1=O)OCCCC3

CHECKPOINT

1 PTS

C[C@@H]1OC2(OCCCC2)C3(OC1=O)OCCCC3

Then analyze the stereoconformation. There are three chiral centers in this structure, so there should be 4 stereoisomers without considering enantiomers.



When O is located at the X_1 or X_2 position, its endocyclic anomeric effect is equivalent.

CHECKPOINT

1 PTS

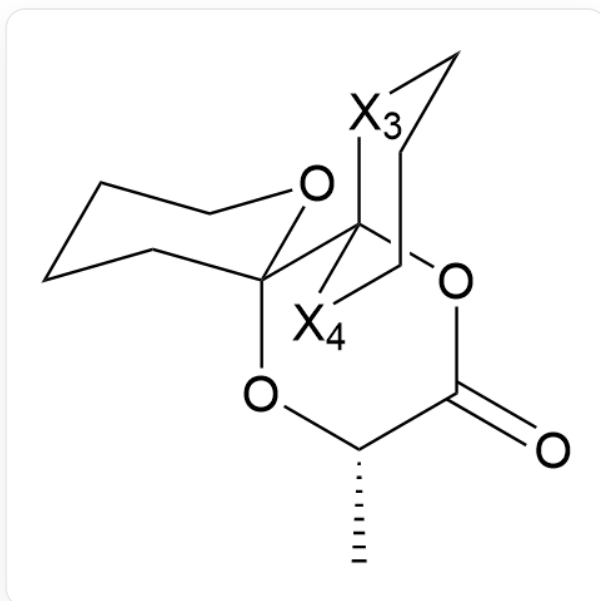
When O is located at the X_1 or X_2 position, its endocyclic anomeric effect is equivalent.

But the difference is that when O is located at the X_1 position, the lone pair electrons of the upright bond O can fill into the antibonding orbital of the $C - X_1$ bond, reducing the system energy through the exocyclic anomeric effect.

CHECKPOINT

1 PTS

But the difference is that when O is located at the X_1 position, the lone pair electrons of the upright bond O can fill into the antibonding orbital of the $C - X_1$ bond, reducing the system energy through the exocyclic anomeric effect.

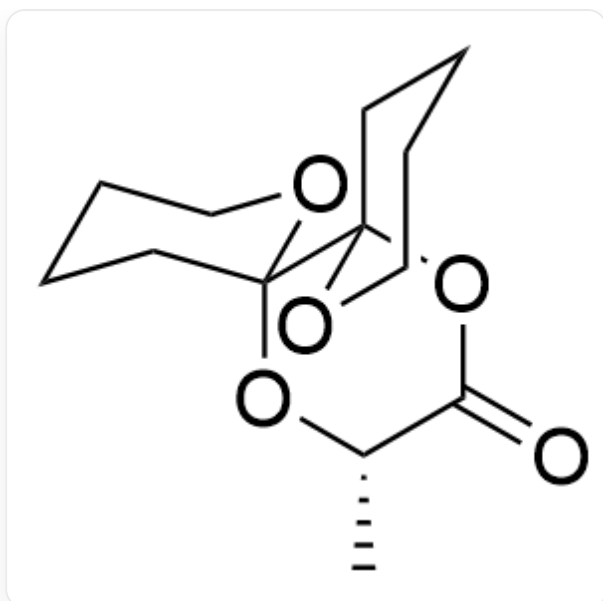


Similarly, O at the X_4 position has a set of exocyclic anomeric effects more than at the X_3 position.

CHECKPOINT

1 PTS

Similarly, O at the X_4 position has a set of exocyclic anomeric effects more than at the X_3 position



O=C(O[C@]12OCCCC2)[C@H](C)O[C@]31CCCCO3

CHECKPOINT

1 PTS

Final product: O=C(O[C@]12OCCCC2)[C@H](C)O[C@]31CCCCO3